

## BOW ISLAND IRRIGATION AGDM, AB, Canada

WMO: 715190

Lat: 49.8684N

Lon: 111.3664W

Elev: 2598

StdP: 13.37

Time Zone: -7.00 (NAM)

Period: 04-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | -17.9      | -12.0 | -23.8                         | 1.6 | -17.5 | -17.9 | 2.3 | -11.9 | 24.0                  | 24.7 | 21.5 | 21.5 | 5.4                   | 190  | 0.617 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 26.9                   | 89.8            | 63.0 | 86.3 | 62.1 | 82.9 | 61.1 | 66.8                | 81.7 | 64.8 | 80.3 | 63.0 | 78.4 | 6.8                  | 250  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 61.7                            | 90.5 | 73.0 | 59.3 | 82.9 | 70.8 | 57.2 | 77.0 | 68.7 | 32.8          | 81.4 | 31.2 | 80.3 | 29.8 | 78.4 | 75.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |       |            |       |    |    |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|-------|------------|-------|----|----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |       | n=50 years |       |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max   | Min        | Max   | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)   | (o)        | (p)   |    |    |
| (4) 21.1          | 18.7 | 16.5 | -28.4                      | 95.9 | 6.5                | 3.0 | -33.0                                              | 98.0 | -36.8      | 99.8 | -40.4      | 101.5 | -45.1      | 103.6 |    |    |
| (5)               |      |      | -28.6                      | 71.0 | 6.4                | 2.8 | -33.2                                              | 73.0 | -36.9      | 74.6 | -40.5      | 76.2  | -45.2      | 78.3  |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 42.5   | 20.7  | 18.3  | 31.5  | 42.8  | 53.4  | 60.9  | 67.2  | 64.9  | 55.9  | 43.5  | 30.3  | 18.8  |
|                                                                                  | DBStd       | 20.59  | 14.82 | 15.48 | 13.67 | 9.07  | 8.04  | 6.04  | 5.34  | 6.25  | 8.40  | 9.25  | 13.57 | 14.92 |
|                                                                                  | HDD50       | 4509   | 909   | 887   | 576   | 242   | 56    | 2     | 0     | 0     | 35    | 240   | 596   | 966   |
|                                                                                  | HDD65       | 8453   | 1374  | 1307  | 1039  | 665   | 366   | 152   | 38    | 80    | 292   | 668   | 1041  | 1431  |
|                                                                                  | CDD50       | 1772   | 0     | 0     | 2     | 28    | 162   | 330   | 533   | 463   | 212   | 38    | 4     | 0     |
|                                                                                  | CDD65       | 241    | 0     | 0     | 0     | 0     | 8     | 30    | 106   | 78    | 19    | 0     | 0     | 0     |
|                                                                                  | CDH74       | 3943   | 0     | 0     | 0     | 18    | 227   | 459   | 1552  | 1251  | 422   | 14    | 0     | 0     |
|                                                                                  | CDH80       | 1369   | 0     | 0     | 0     | 2     | 50    | 119   | 583   | 472   | 143   | 0     | 0     | 0     |
| Wind                                                                             | WSAvg       | 7.7    | 8.5   | 7.8   | 8.6   | 8.8   | 8.2   | 7.3   | 6.2   | 6.2   | 6.9   | 7.9   | 8.1   | 8.0   |
| Precipitation                                                                    | PrecAvg     | 14.3   | 0.7   | 0.5   | 0.7   | 1.2   | 1.9   | 3.3   | 1.4   | 1.3   | 1.2   | 0.8   | 0.6   | 0.6   |
|                                                                                  | PrecMax     | 20.9   | 1.7   | 1.1   | 2.0   | 2.7   | 4.0   | 8.1   | 4.9   | 3.1   | 2.7   | 2.8   | 1.3   | 1.3   |
|                                                                                  | PrecMin     | 7.6    | 0.2   | 0.1   | 0.0   | 0.3   | 0.3   | 0.6   | 0.1   | 0.1   | 0.0   | 0.2   | 0.1   | 0.0   |
|                                                                                  | PrecStd     | 3.0    | 0.4   | 0.3   | 0.5   | 0.6   | 0.9   | 1.6   | 1.1   | 0.8   | 0.8   | 0.6   | 0.4   | 0.3   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 51.6  | 51.8  | 64.9  | 76.3  | 85.4  | 89.0  | 94.0  | 92.7  | 90.4  | 76.0  | 64.0  | 51.3  |
|                                                                                  |             | MCWB   | 41.8  | 40.7  | 48.0  | 52.0  | 58.3  | 64.1  | 64.1  | 62.6  | 60.3  | 55.0  | 48.0  | 40.7  |
|                                                                                  | 2%          | DB     | 45.8  | 45.6  | 59.4  | 69.4  | 80.2  | 83.3  | 90.1  | 89.3  | 84.3  | 68.7  | 56.2  | 45.0  |
|                                                                                  |             | MCWB   | 38.3  | 38.1  | 45.3  | 49.8  | 56.4  | 61.5  | 64.9  | 61.5  | 58.6  | 52.0  | 44.5  | 38.2  |
|                                                                                  | 5%          | DB     | 42.5  | 41.7  | 54.4  | 64.4  | 75.4  | 79.3  | 86.4  | 85.5  | 78.7  | 63.2  | 51.5  | 41.7  |
|                                                                                  |             | MCWB   | 36.6  | 35.8  | 42.8  | 47.1  | 54.3  | 60.0  | 64.0  | 61.1  | 57.0  | 49.2  | 42.3  | 36.1  |
|                                                                                  | 10%         | DB     | 39.8  | 38.4  | 49.3  | 59.4  | 70.5  | 75.3  | 82.8  | 81.5  | 73.2  | 58.6  | 47.1  | 38.6  |
|                                                                                  |             | MCWB   | 35.1  | 33.8  | 39.9  | 44.4  | 52.1  | 58.2  | 62.9  | 60.4  | 54.8  | 46.8  | 39.1  | 33.7  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 42.2  | 42.8  | 49.0  | 53.4  | 59.5  | 67.5  | 71.2  | 66.9  | 63.1  | 56.3  | 50.0  | 42.3  |
|                                                                                  |             | MCDB   | 50.1  | 50.1  | 62.0  | 71.8  | 81.0  | 80.0  | 84.7  | 82.9  | 79.4  | 72.3  | 61.8  | 49.4  |
|                                                                                  | 2%          | WB     | 38.9  | 38.6  | 46.4  | 50.7  | 57.3  | 64.1  | 68.2  | 64.7  | 60.2  | 52.9  | 45.7  | 38.6  |
|                                                                                  |             | MCDB   | 45.2  | 44.8  | 57.4  | 68.1  | 76.4  | 79.4  | 82.2  | 81.5  | 80.1  | 66.0  | 55.6  | 43.7  |
|                                                                                  | 5%          | WB     | 36.9  | 36.0  | 43.3  | 47.9  | 55.4  | 61.6  | 66.1  | 63.0  | 57.9  | 50.1  | 42.6  | 36.4  |
|                                                                                  |             | MCDB   | 41.8  | 41.1  | 53.4  | 62.0  | 72.1  | 75.8  | 81.3  | 79.8  | 76.0  | 61.1  | 50.5  | 41.6  |
|                                                                                  | 10%         | WB     | 35.1  | 33.9  | 40.2  | 45.5  | 53.5  | 59.5  | 64.3  | 61.5  | 55.7  | 47.6  | 39.9  | 33.8  |
|                                                                                  |             | MCDB   | 39.7  | 38.3  | 48.5  | 57.7  | 67.5  | 72.4  | 79.0  | 77.5  | 72.2  | 57.9  | 46.5  | 38.4  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 18.2  | 19.1  | 19.9  | 22.7  | 23.7  | 22.7  | 26.9  | 27.1  | 25.7  | 22.2  | 19.5  | 17.2  |
|                                                                                  |             | MCDBR  | 18.2  | 21.9  | 26.9  | 32.8  | 32.4  | 29.2  | 33.2  | 34.8  | 35.8  | 28.8  | 24.5  | 18.6  |
|                                                                                  |             | MCWBR  | 12.9  | 15.8  | 15.5  | 16.4  | 13.3  | 12.6  | 13.8  | 13.2  | 14.6  | 14.8  | 15.0  | 13.1  |
|                                                                                  | 5% WB       | MCDBR  | 18.2  | 21.0  | 25.9  | 29.8  | 28.4  | 27.0  | 29.9  | 30.0  | 34.1  | 27.1  | 23.8  | 19.3  |
|                                                                                  |             | MCWBR  | 13.3  | 15.7  | 15.5  | 15.6  | 12.4  | 12.9  | 13.7  | 12.5  | 14.4  | 14.5  | 15.2  | 14.3  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.268  | 0.271 | 0.295 | 0.346 | 0.368 | 0.359 | 0.358 | 0.380 | 0.337 | 0.295 | 0.265 | 0.257 |       |
|                                                                                  | taud        | 2.416  | 2.471 | 2.463 | 2.315 | 2.299 | 2.380 | 2.397 | 2.306 | 2.407 | 2.506 | 2.493 | 2.411 |       |
|                                                                                  | Ebn at Noon | 252    | 283   | 293   | 285   | 282   | 285   | 283   | 271   | 273   | 269   | 253   | 240   |       |
|                                                                                  | Edh at Noon | 19     | 23    | 29    | 37    | 39    | 36    | 36    | 37    | 30    | 23    | 18    | 17    |       |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 397    | 711   | 1133  | 1567  | 1857  | 2033  | 2212  | 1780  | 1269  | 789   | 442   | 309   |       |
|                                                                                  | RadStd      | 36     | 59    | 79    | 109   | 133   | 120   | 111   | 113   | 104   | 79    | 31    | 27    |       |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (63 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page